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HW 2

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Table 1 MSE and PSNR of the Reconstructed Luminance Component

Introduction

Storing high-resolution images requires substantial memory and bandwidth. Image compression reduces redundancy and allows efficient storage and transmission without significant visual loss.

This project implements **JPEG-style compression** using the 4:2:0 YCbCr format. The workflow includes:

- RGB $\rightarrow$ YCbCr conversion
- 4:2:0 chroma subsampling
- 8 $\times$ 8 block DCT transformation
- Quantization using JPEG luminance and chrominance matrices
- Zigzag scanning of coefficients
- Inverse quantization and inverse DCT
- Image reconstruction

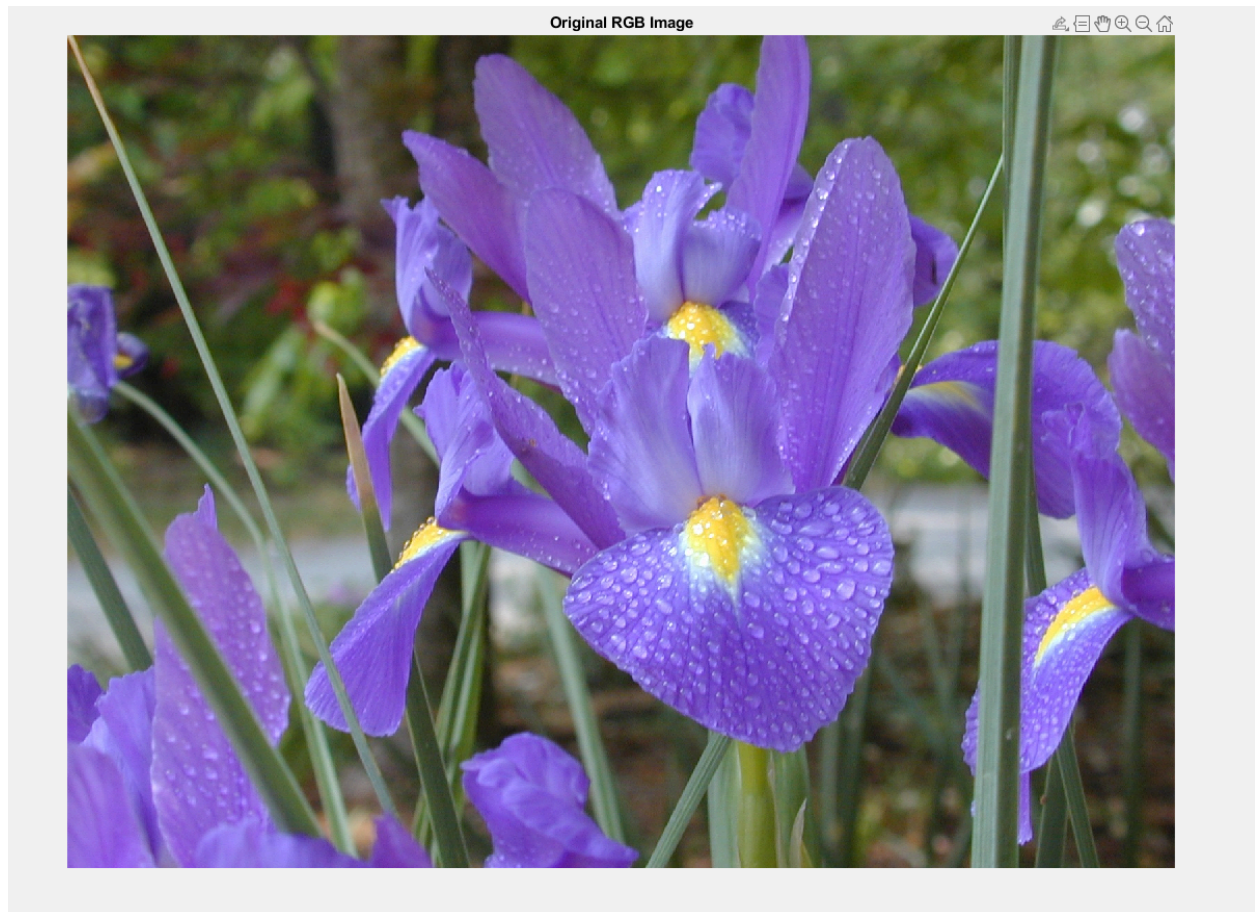
Quality evaluation is performed via **PSNR**, **MSE**, and visual comparison between the original and reconstructed images through an **Error Image**.

Procedural Section

Input (Figure 1)

The program reads **Iris.jpg** and converts it to double precision for processing.

Figure 1: Original Image



Color Space Conversion and Subsampling

- Convert to YCbCr.
- Separate Y, Cb, Cr bands.
- Perform **4:2:0 subsampling**: retain every other row and column of Cb and Cr.

DCT Transformation (Figures 2 & 3)

- Divide Y, Cb, Cr bands into **8x8 blocks**.
- Compute **2D DCT** using MATLAB **dct2**.
- Display **DCT matrix** of the Y component.
- Show **88th and 89th blocks of the 63rd row** of DCT coefficients for luminance.

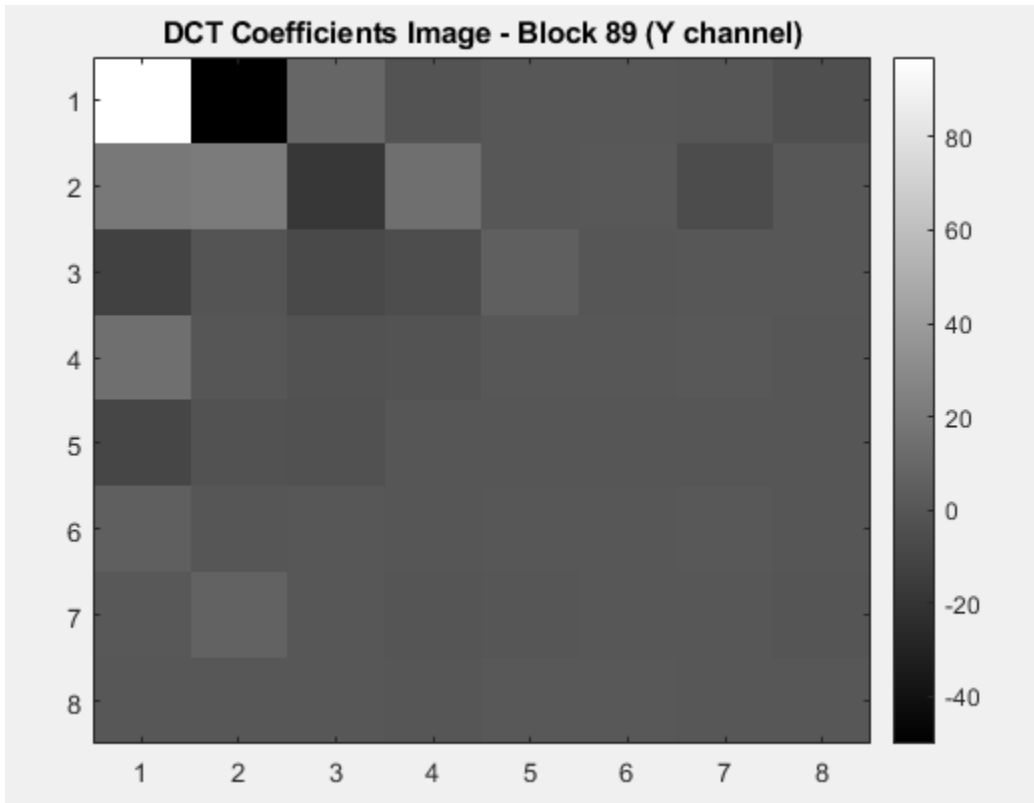
Figure 2: DCT Coefficient Matrix of Y Component

DCT Coefficients for Block 88 (63rd row):

14.2500	-7.7117	-1.2901	-2.9340	0	0.2066	0.8050	-0.0574
85.5393	26.7868	-0.5195	2.0161	-0.2250	0.0842	-0.2657	0.5718
-43.1998	-5.4007	0.3018	0.0345	-0.4619	-0.1734	-0.4786	-0.1037
13.5677	-1.6087	-3.6478	0.0126	-0.1274	-0.8315	0.8921	-0.3825
-8.7500	2.6183	-0.2870	0.0153	0.5000	0.4977	-0.6929	-0.1474
4.5165	-3.7514	0.1975	0.4488	0.6407	-0.1244	0.1556	0.6288
-5.6481	-0.1470	-0.2286	0.0748	0.1913	0.0639	-0.0518	-0.0982
-0.5400	0.3521	-0.5339	-0.3217	-0.1503	-0.3500	0.6773	0.3250

DCT Coefficients for Block 89 (63rd row):

97.1250	-50.1109	9.0290	-2.0219	0.1250	0.2191	-0.0869	-4.6714
19.3343	21.1792	-17.9763	14.0860	0.3760	0.5391	-6.0568	0.4348
-12.2230	-1.3664	-7.9775	-5.5110	4.6638	-0.4163	-0.0366	0.2563
13.8557	-0.4318	-2.5980	-2.3671	0.3170	-0.0551	0.6896	-0.4083
-9.6250	-2.4023	-3.1823	-0.2462	-0.1250	-0.3771	-0.1701	-0.2976
4.8984	-0.2277	0.2143	-0.2319	-0.0631	0.0230	0.7130	-0.1818
0.8687	6.6496	0.2134	-1.0011	-0.1729	0.0494	-0.0225	-0.6519
0.1498	0.2581	0.2375	-0.2891	0.5627	0.7221	0.4555	0.1649



Quantization (Figure 4)

- Use JPEG **luminance** and **chrominance** quantization matrices.
- Divide each DCT coefficient by the quantization matrix and round.
- Apply **zigzag scanning**.

Figure 4: Zigzag Scanned DCT Coefficients

Block 88:

[illegible]

Block 89:

$$[6, -1, 1, 2, 0, 0, 0, 0, 2, -1, -5, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, -1, 0, 0, 0, 0, 0, 0, 1, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0]$$

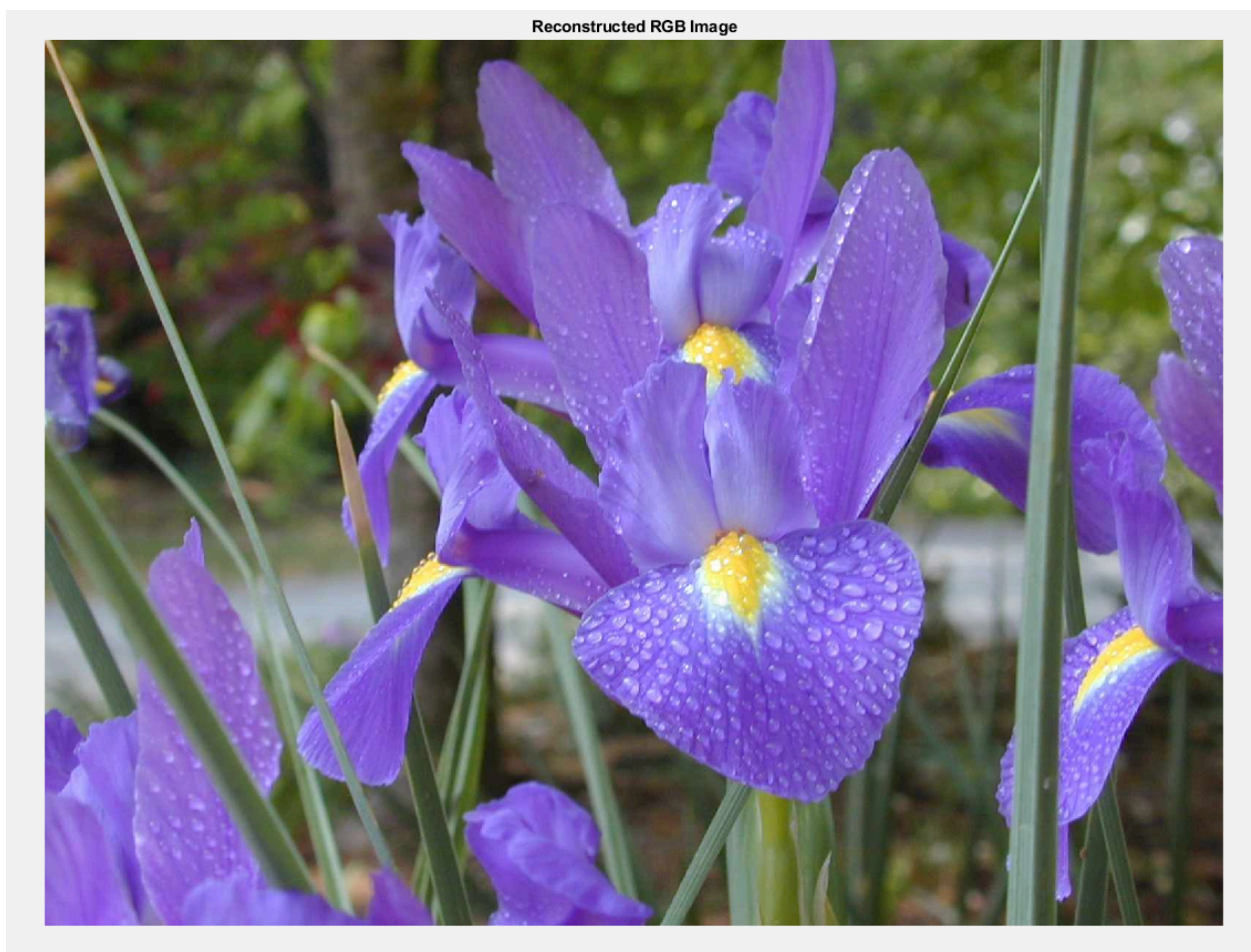
Inverse Quantization

- Multiply quantized DCT coefficients by the quantization matrix to approximate original DCT coefficients.

Inverse DCT and Reconstruction (Figure 5)

- Apply **inverse DCT** (`idct2`) to each block.
- Reassemble Y, Cb, Cr bands.
- Upsample chrominance bands to match Y.
- Convert reconstructed YCbCr back to RGB.

Figure 5: Reconstructed RGB Image



Error Metrics (Table 1 and Figure 6)

Compute **MSE** and **PSNR** for Y component of Original vs. Reconstructed Image:

Table 1: MSE and PSNR of Luminance Component

Metric	Value
MSE	6.42
PSNR	40.06 dB

Figure 6: Error Image (Original – Reconstructed, Y Component)



Results

- **DCT Coefficients:** Most energy concentrated in low frequencies.
- **Zigzag coefficients:** Show efficient energy compaction, zeros for most of the right-hand side of the vector – run length code will be effective in compacting the zeros.
- **Reconstructed Image:** Minimal visual difference from original.
- **Error Image:** Errors mostly in high-frequency areas such as edges, hardly noticeable for the given image and next to no difference as shown by an almost entirely black image (difference value of 0).
- **PSNR:** A value of 40.06dB represents a high reconstruction quality, confirming minimal visual loss between the original and reconstructed image.

Conclusion

- JPEG-style compression effectively reduces data while maintaining visual fidelity.
- **4:2:0 chroma subsampling** achieves 2:1 compression with minimal perceptual loss.
- **DCT-based quantization** compacts energy into fewer coefficients.
- Low **MSE** and high **PSNR** confirm high-quality reconstruction.
- This workflow demonstrates practical methods used in image and video compression standards.